

Disclaimer

Thank you for your interest in using Google Cloud training material. We are excited about the content we are able to provide you (collectively, “Teaching Resources”) and hope you are too.

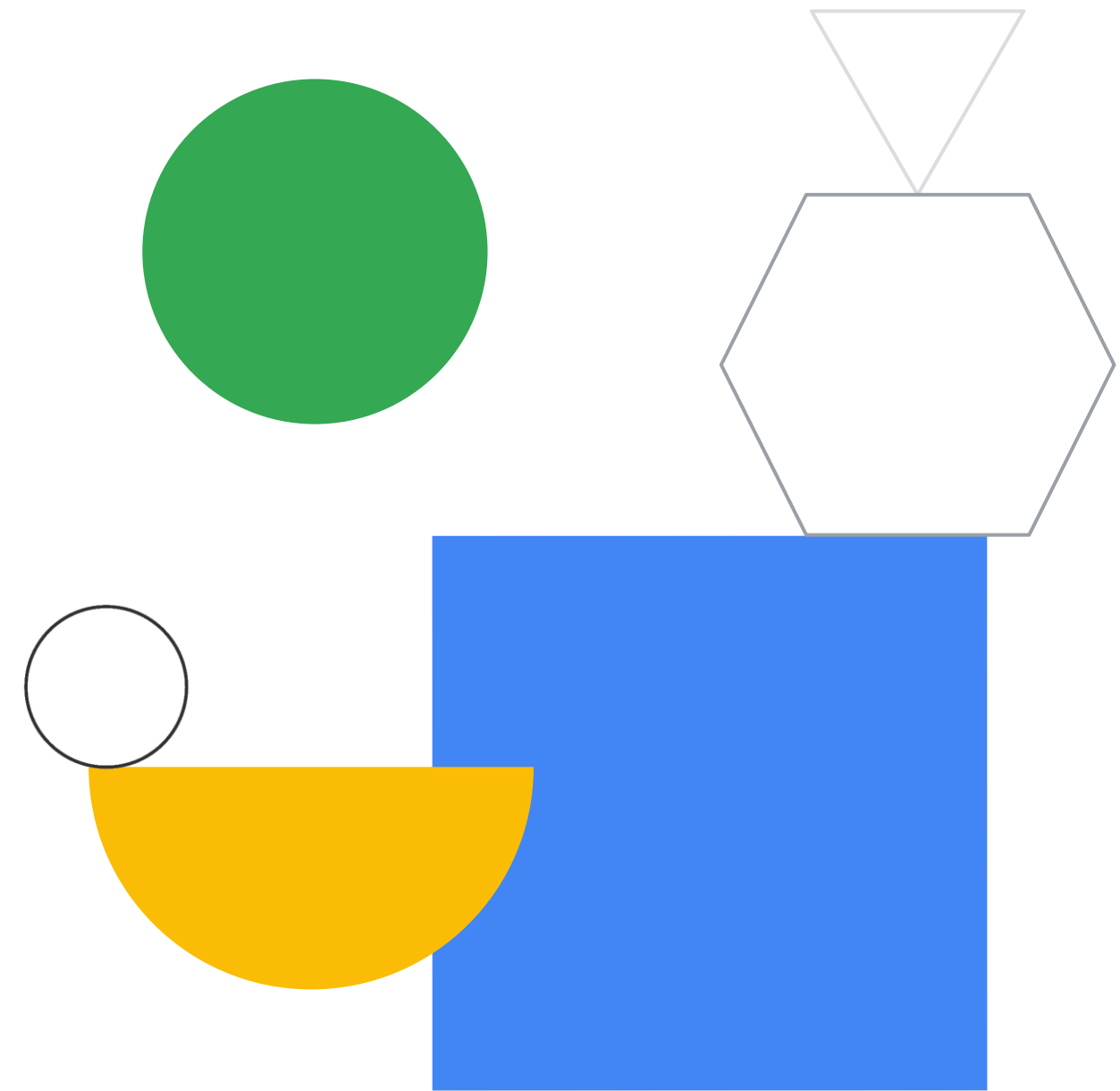
By using Teaching Resources, you agree to be bound by all of the following terms, as well as the [Google terms of service](#) and the [Google privacy policy](#). Unless otherwise specified, terms used below will have the meanings described in the [Google terms of service](#).

1. **Educational Use Only.** Teaching Resources is intended only for use teaching courses at regionally accredited or higher learning institutions. The content may be adapted, customize, remixed, and shared for educational use. Content may not be distributed, or otherwise exploited for any commercial purpose, commercial advantage, or private monetary compensation.
1. **Attribution Requirements.** If you distribute, publicly perform, display, transmit, publish, or otherwise make available any Teaching Resources or any derivative works thereof, you must attribute the material you use back to the Teaching Resources, but not in any way that suggests Google, any of its affiliates, or any of its third party content providers endorse you or your use of such materials. If you adapt, remix, or customize the Teaching Resources please include the following text on each edited slide: *“The original content was provided by Google LLC and modified for the purpose of the course, without input or endorsement from Google LLC.”*
1. Descriptions of Google products, services, infrastructure and processes contained in the Teaching Resources are descriptions for teaching and learning purposes only and do not constitute a guarantee, promise, or representation of accuracy by Google. Pricing, availability or features of Google Cloud products and services described in the Teaching Resources may change.

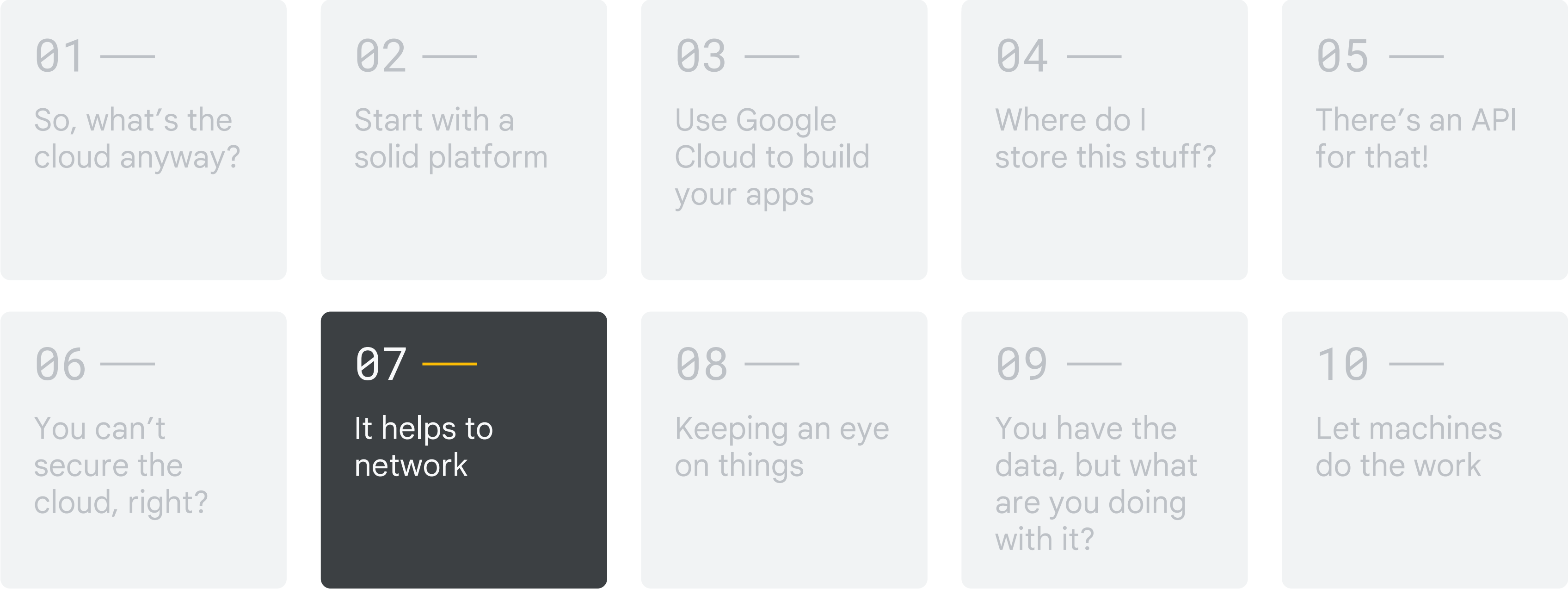


It helps to network

Module 7 - Networks
Google Cloud Computing Foundations



Course map



Google Cloud skill badges

Objectives

- 01 Explore Virtual Private Clouds (VPCs).
- 02 Examine Google's network architecture.
- 03 Learn how to use multiple VPC networks.
- 04 Explore options to build hybrid clouds.
- 05 Examine load balancing options.





Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

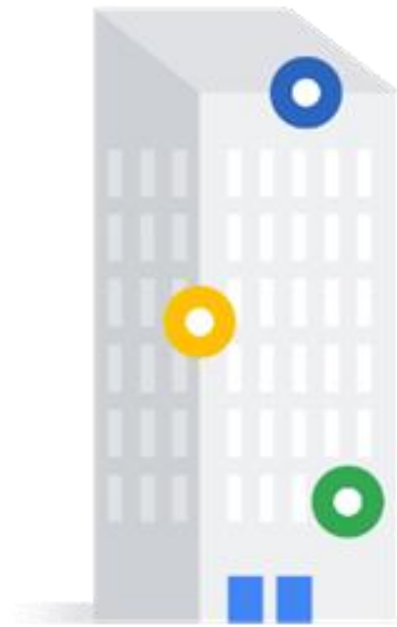


Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

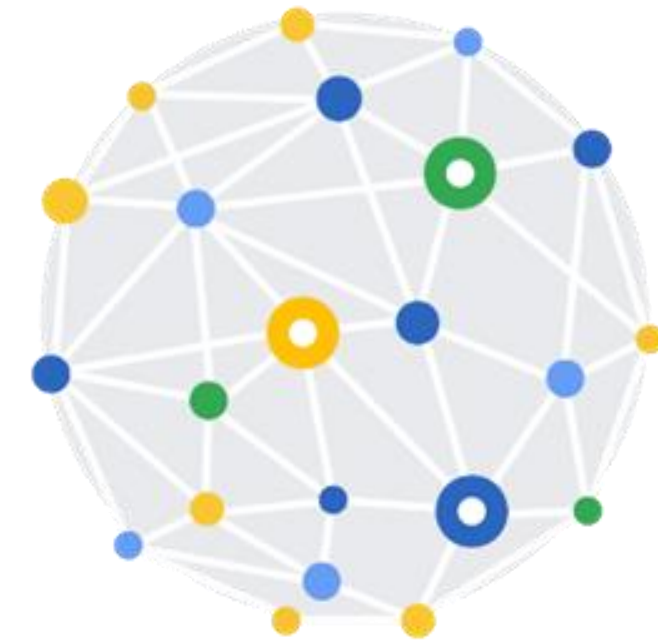
A background in networking



Local area network
(LAN)

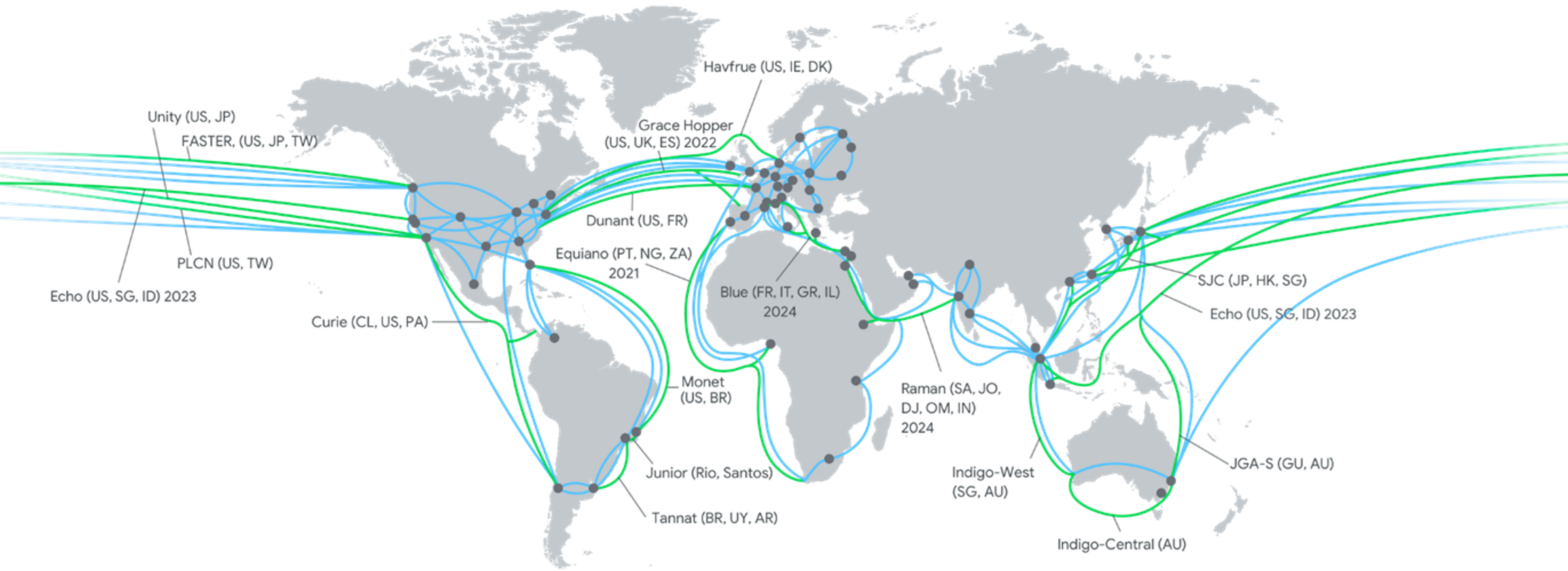


Wide area network
(WAN)



Internet

Google Cloud PoPs and Network





Agenda

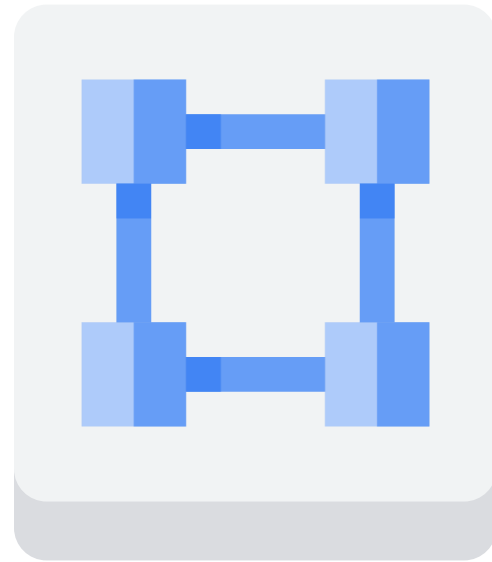


- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)**
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

Virtual Private Cloud is your cloud within the Cloud



A **Virtual Private Cloud**, or **VPC**, is a secure, individual, private cloud-computing model hosted within a public cloud



Virtual Private Cloud



VPC networks connect Google Cloud resources to each other and to the internet



Segmenting networks



Using firewall rules to restrict access to instances

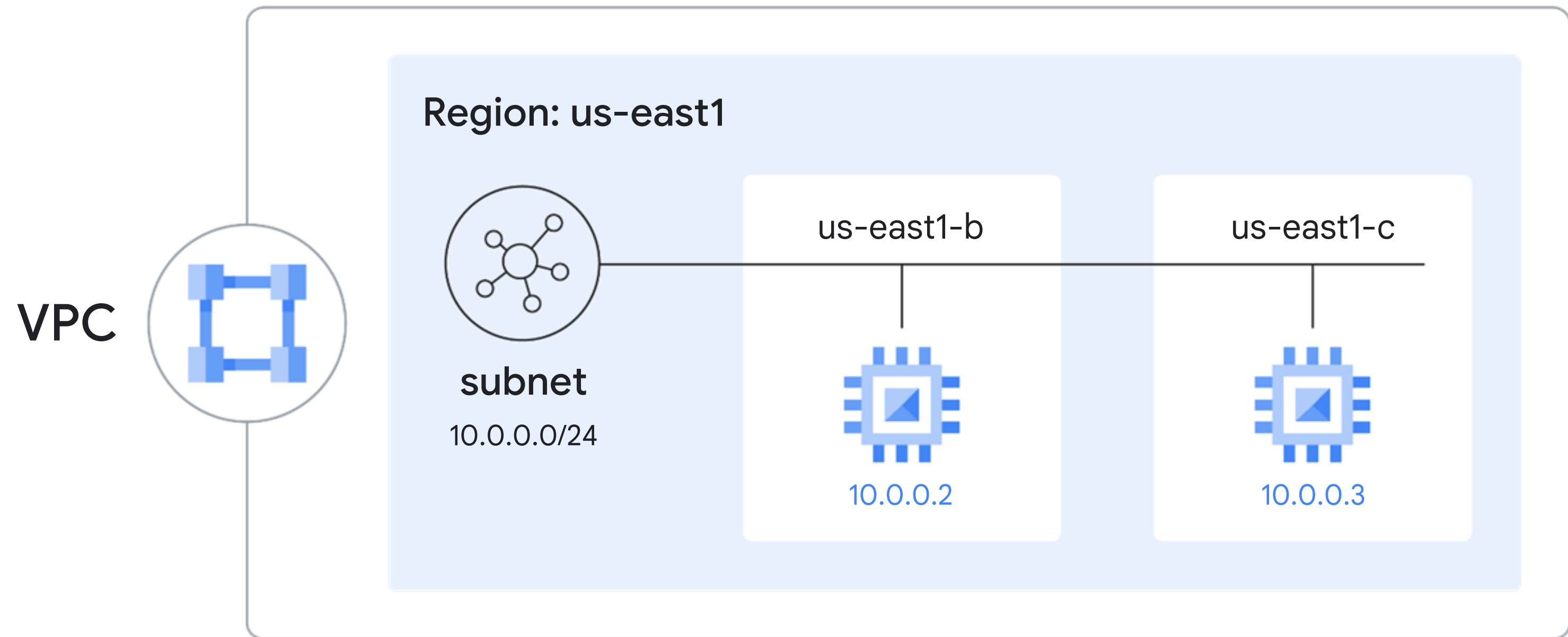


Creating static routes to forward traffic to specific destinations



Google VPC networks are global and can have subnets in any Google Cloud region worldwide

VPC subnets connect resources in different zones



Auto and custom networks



Auto Mode

- Default network
- One subnet per region
- Regional IP allocation
- Fixed /20 subnetwork per region
- Expandable up to /16
- Suitable for isolated use cases



Custom Mode

- No default subnets created
- Full control of IP ranges
- Regional IP allocation
- Expandable to IP ranges you specify
- Suitable to production environments

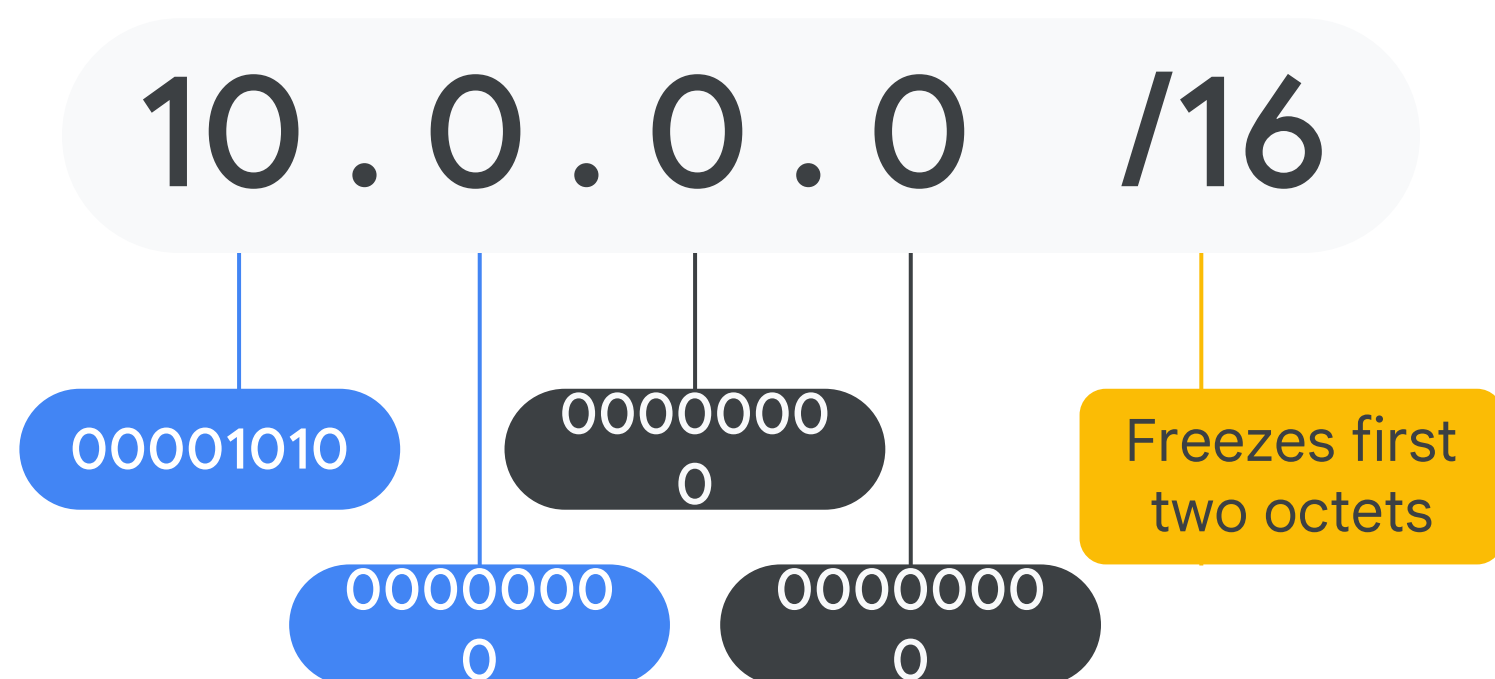


Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses**
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

A VPC is made up of subnets



Subnets need to be configured with a private IP address range.

IP addresses are used for internal network communication.

Each octet is represented by 8 bits.

The `/###` determines the number of address bits that are static

A /16 range will provide 65,536 available IP addresses

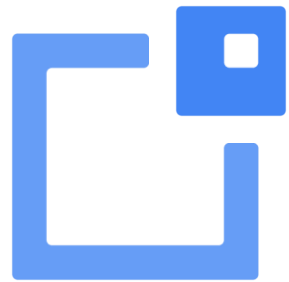
CIDR IP address totals						
/16	/17	/18	/19	/20	/21	/22
65,536	32,768	16,384	8,192	4,096	2,048	1,024
/23	/24	/25	/26	/27	/28	
512	256	128	64	32	16	

The CIDR range determines how many IP addresses are available.

Every time you add “1” to the last number, the number of available IP addresses is cut in half.

The basics of public and internal IP addresses

Cloud External
IP Addresses



The Internet



Virtual machines



Public (external) IP addresses

- Can be assigned from pool (ephemeral) or reserved (static).
- Billed when not attached to a running VM.
- VM doesn't know the external IP; it's mapped to the internal IP.

Private (internal) IP addresses

- Allocated from subnet range to VMs by DHCP.
- DHCP lease is renewed every 24 hours.
- VM name and IP is registered with network-scoped DNS.



Agenda



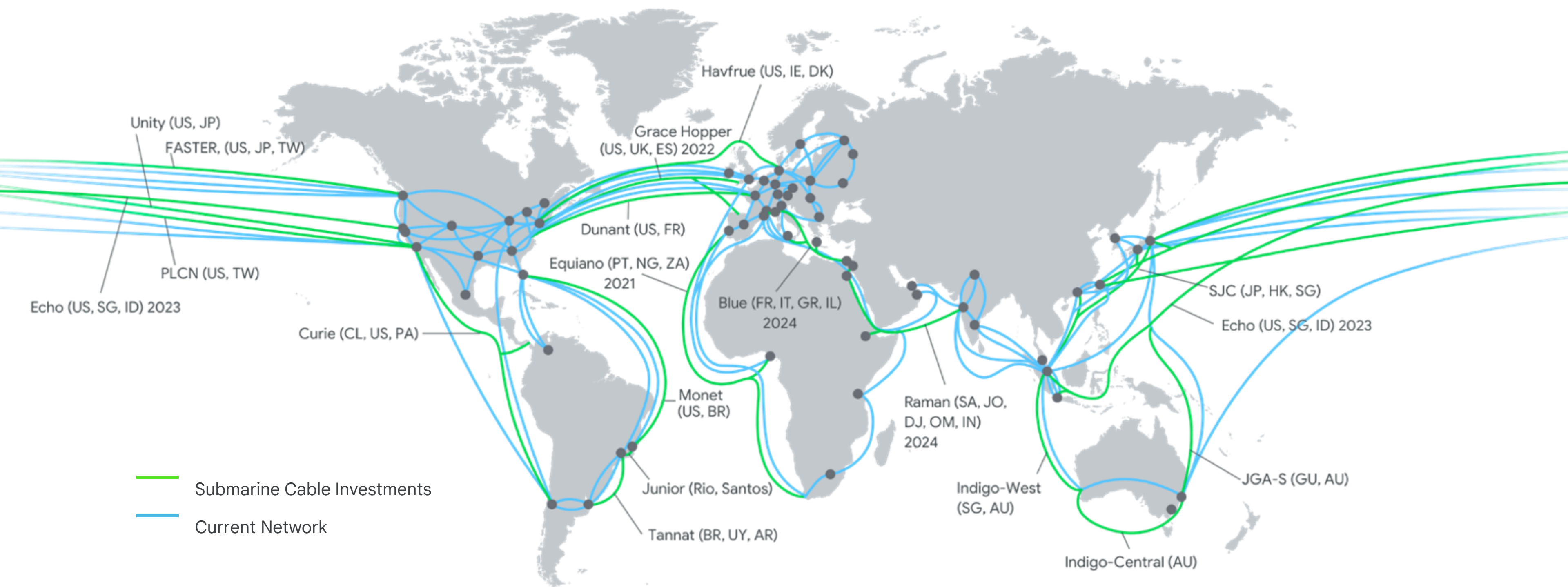
- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network**
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

Google Cloud regions and zones



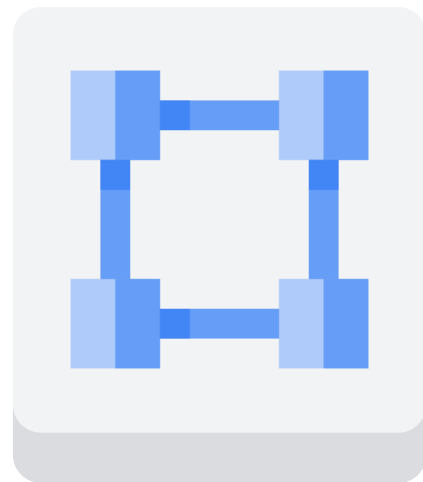
*Exception: region has 4 zones.

Google Cloud regions and PoPs



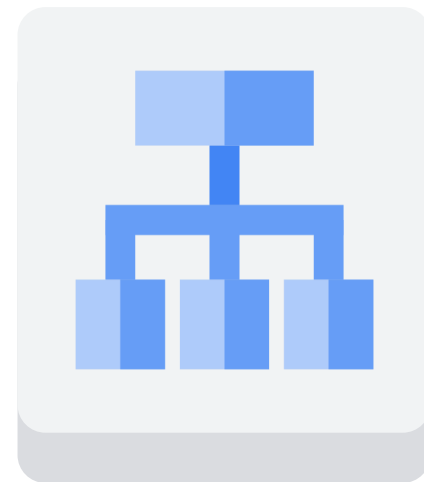
Google Cloud's primary networking products

Google Cloud
VPC



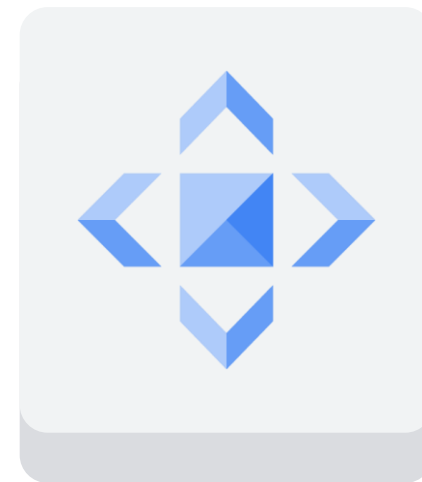
Comprehensive
networking
capabilities and
infrastructure

Cloud Load
Balancing



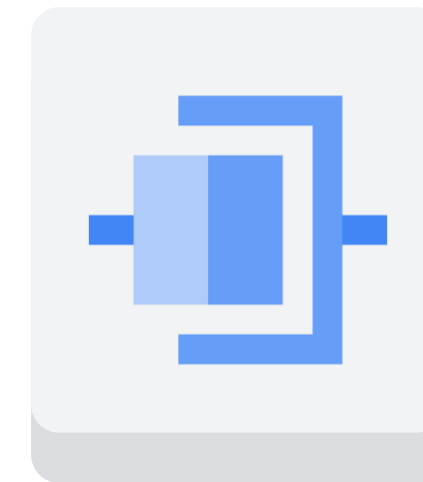
High performance,
scalable load
balancing

Cloud CDN



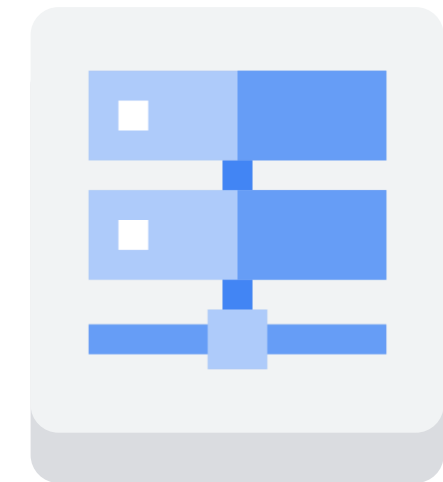
Low-latency, low-
cost content
delivery

Cloud
Interconnect



Fast, high
availability
interconnect

Cloud DNS



Highly available
global DNS
network

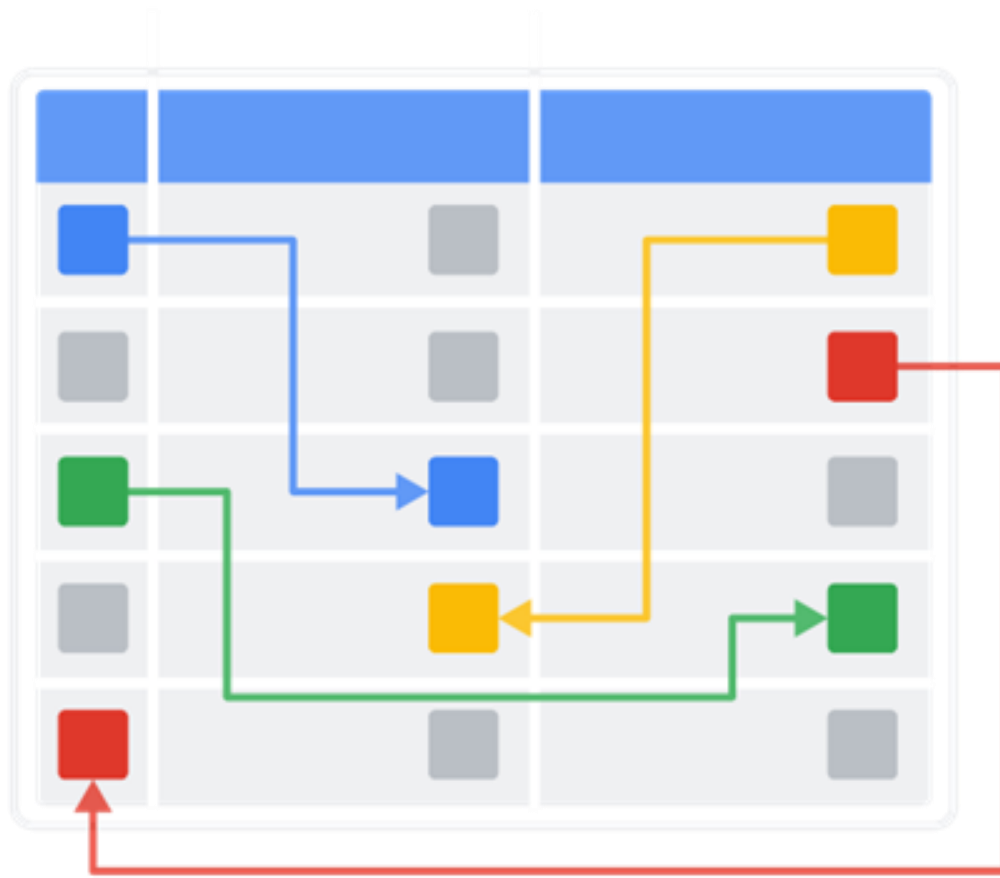


Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud**
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

VPCs do not require a router to be provisioned



Routing tables



Built-in so you don't have to provision or manage a router



Used to forward traffic from one instance to another

VPCs also do not require a firewall to be provisioned



Firewall



Restricts access to instances through both incoming and outgoing traffic



Rules can be defined through metadata tags on Compute Engine instances

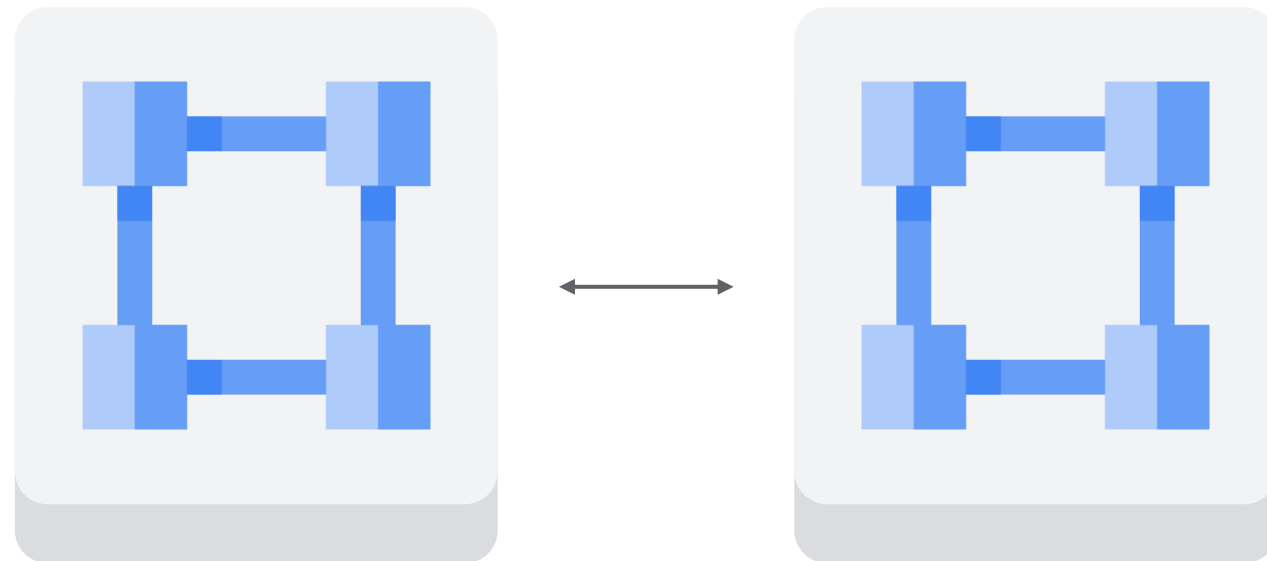


Agenda

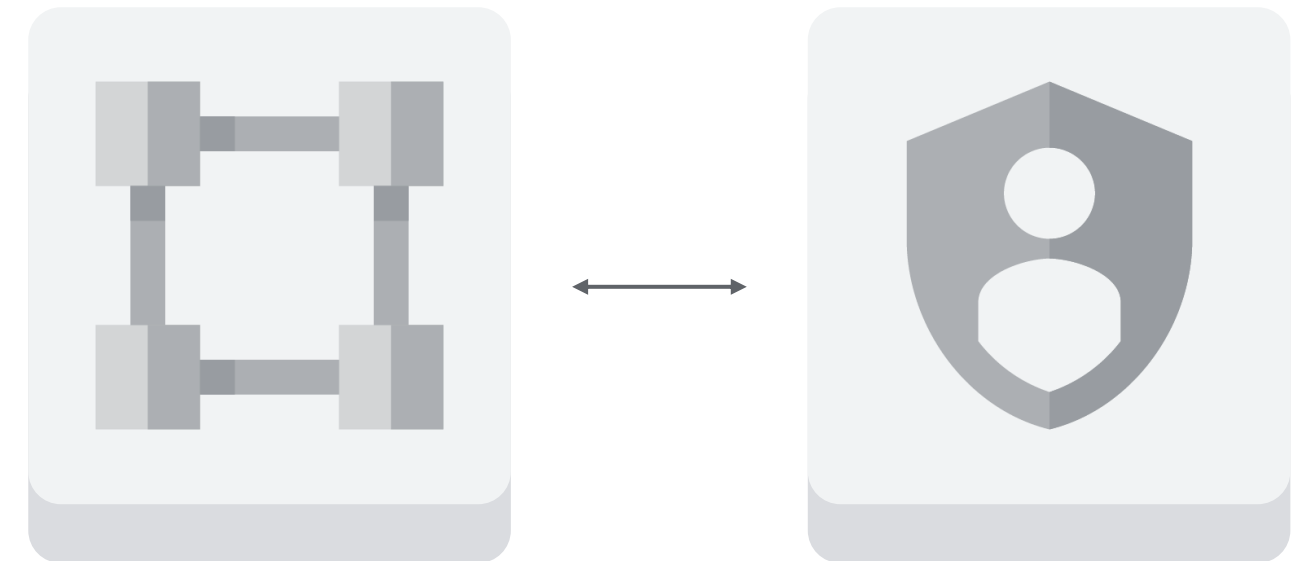


- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks**
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

VPC peering and sharing allow projects to communicate

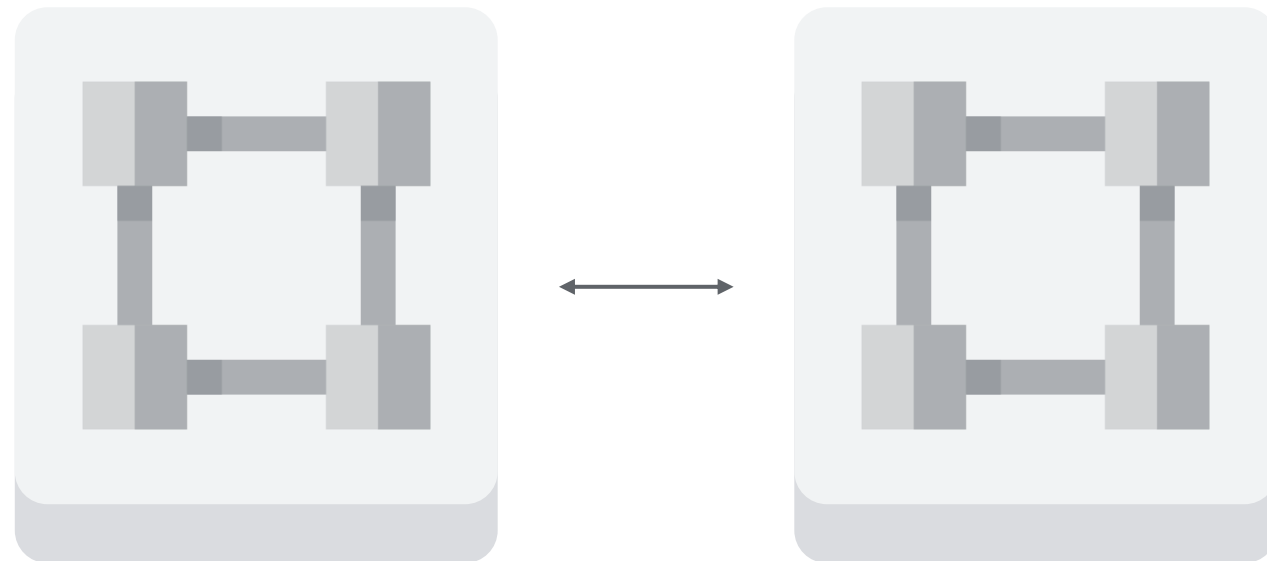


VPC peering

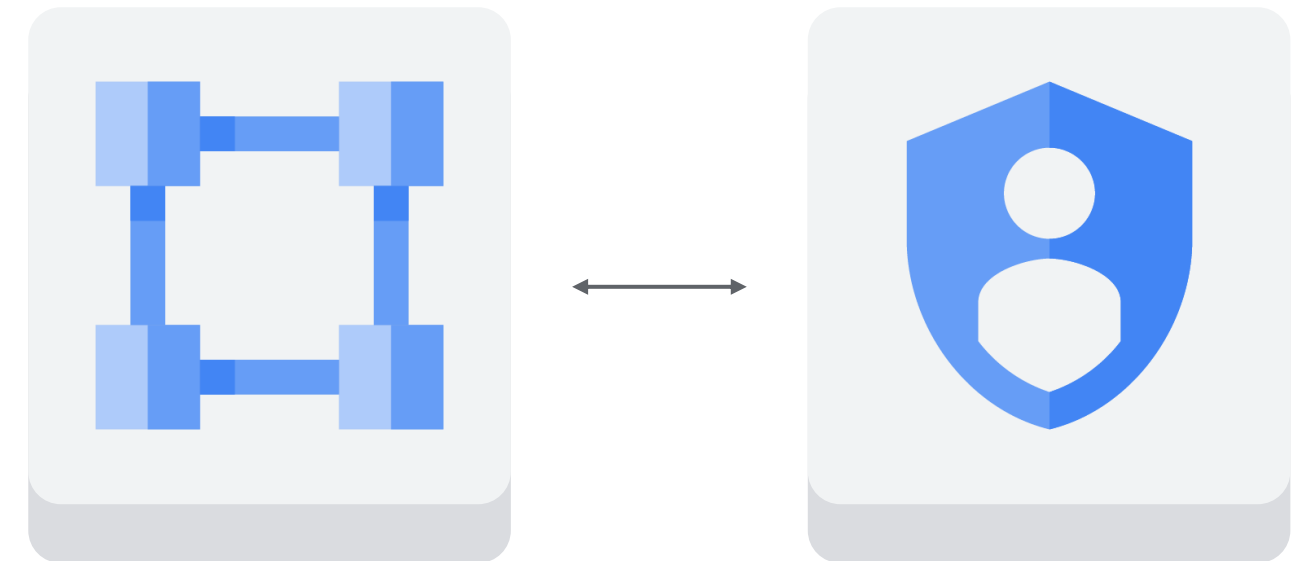


Shared VPC

VPC peering and sharing allow projects to communicate



VPC peering



Shared VPC



Agenda



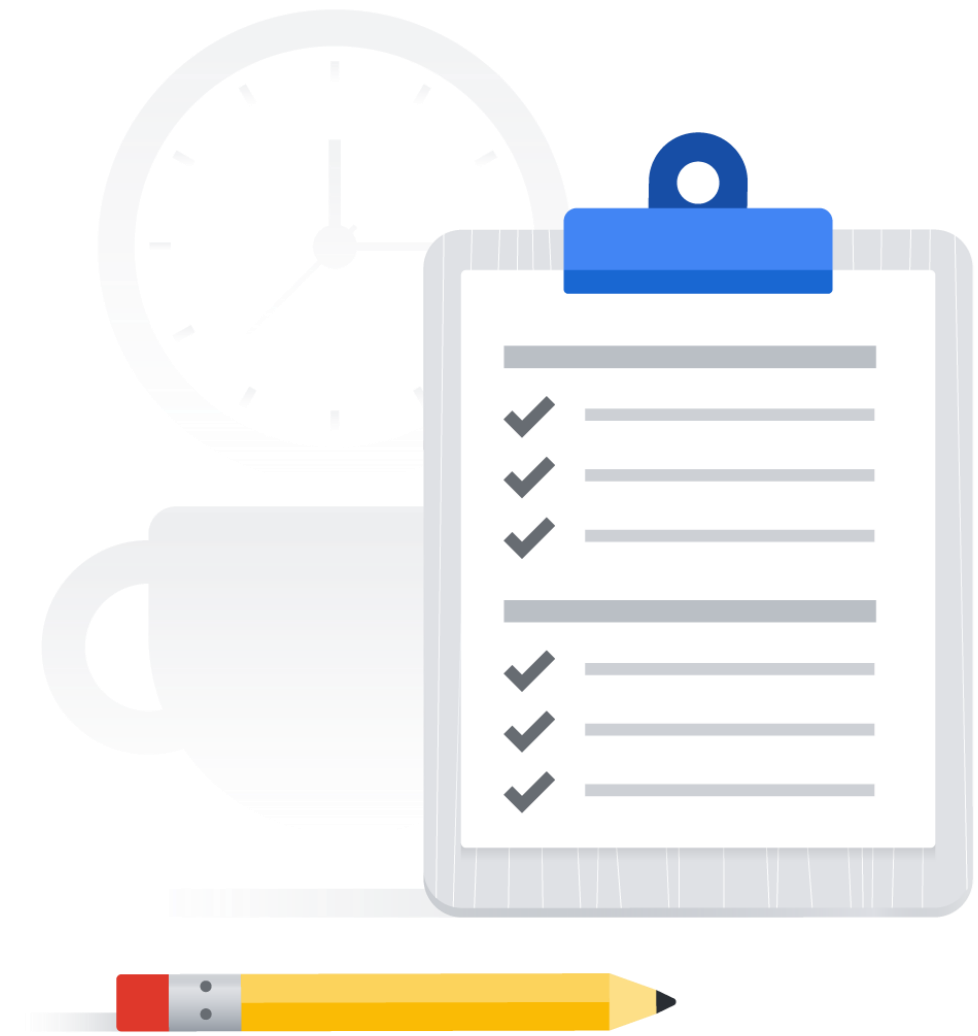
- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks**
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

Lab: Multiple VPC Networks

 Hands-on lab

During this lab, you'll:

1. Create custom mode VPC networks with firewall rules.
2. Use Compute Engine to create virtual machine instances.
3. Explore the connectivity for virtual machine instances across VPC networks.
4. Create a virtual machine instance with multiple network interfaces.



Agenda

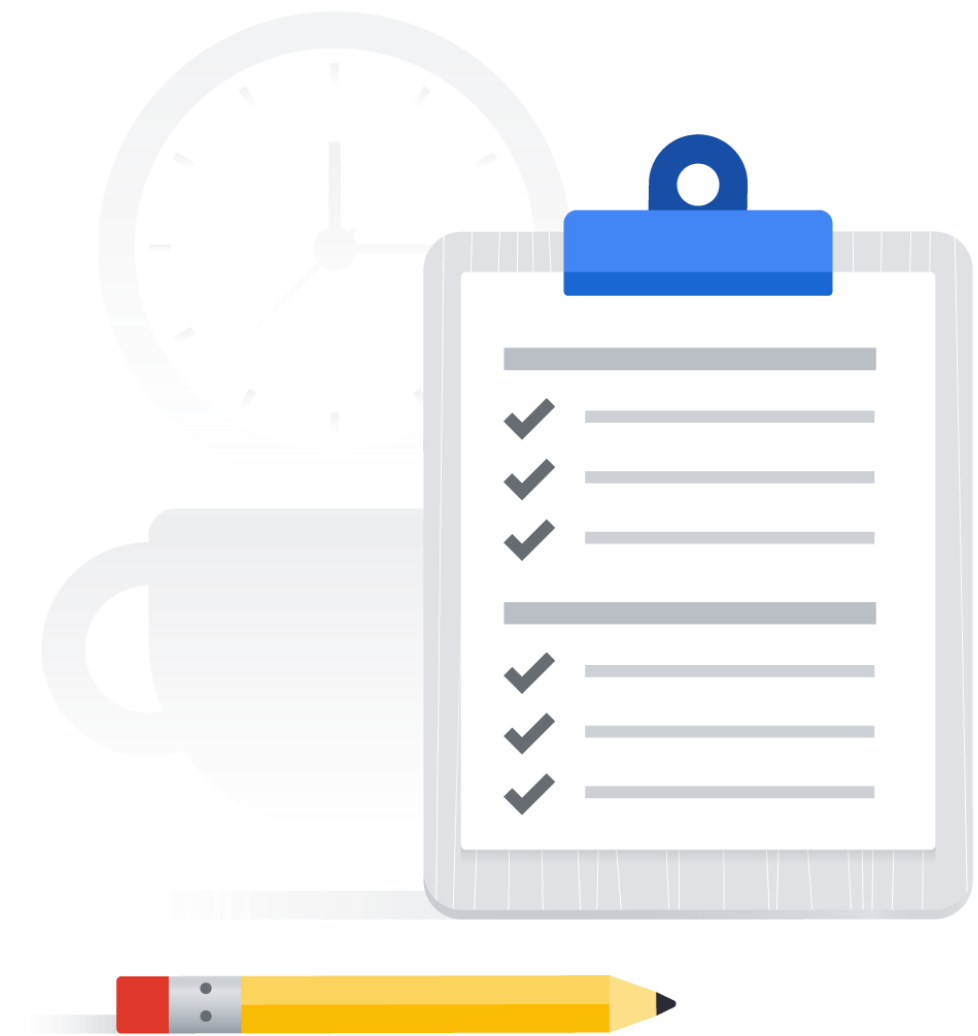
- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access**
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

Lab: VPC Networks - Controlling Access

 Hands-on lab

During this lab, you'll:

1. Create an NGINX web server.
2. Create tagged firewall rules.
3. Create a service account with IAM roles.
4. Explore permissions for the Network Admin and Security Admin roles.



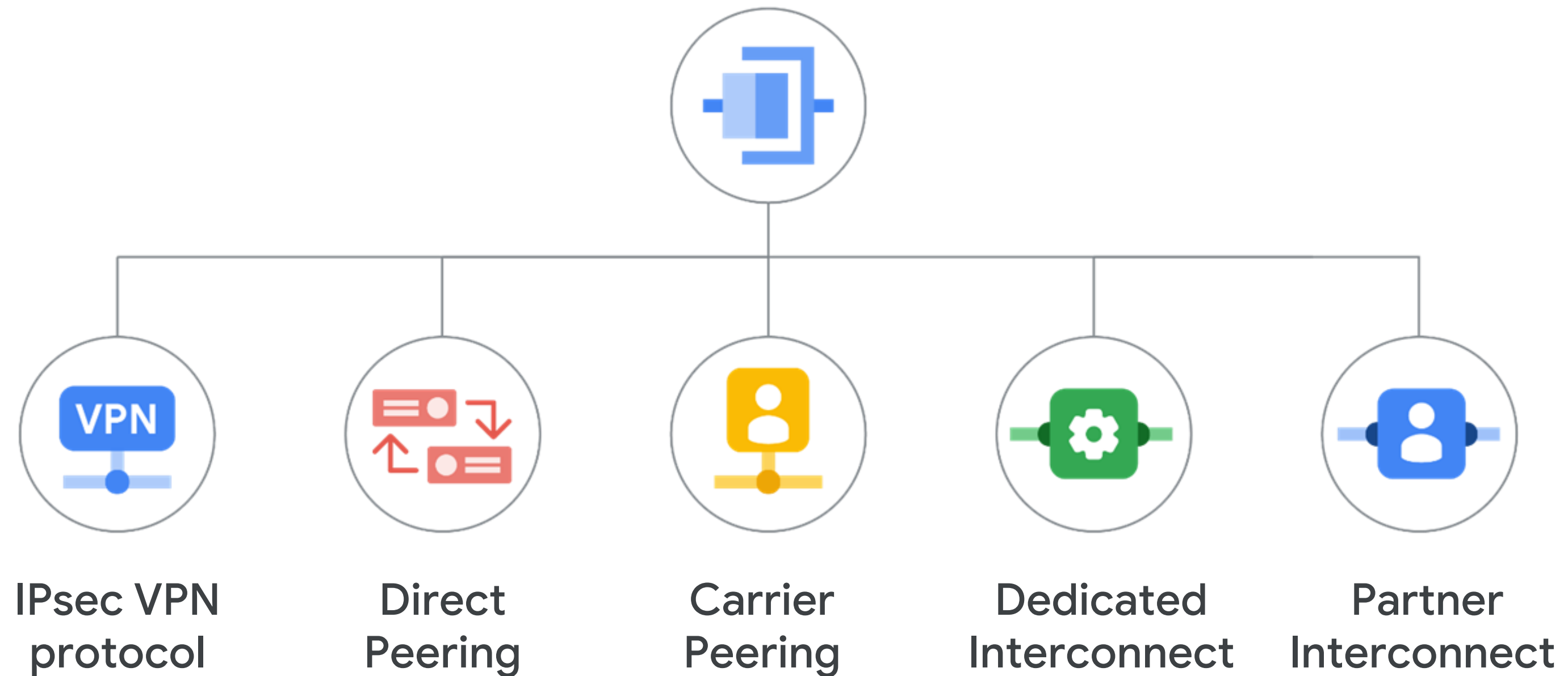


Agenda



- | | |
|----|---|
| 01 | Networking in the cloud |
| 02 | Virtual Private Clouds (VPCs) |
| 03 | The basics of public and internal IP addresses |
| 04 | The Google Cloud network |
| 05 | Routes and firewall rules in the cloud |
| 06 | Multiple VPC networks |
| 07 | Lab: Multiple VPC Networks |
| 08 | Lab: VPC Networks - Controlling Access |
| 09 | Building hybrid clouds |
| 10 | Load balancing options |
| 11 | Lab: HTTP Load Balancer with Google Cloud Armor |
| 12 | Quiz |
| 13 | Summary |

Google VPNs can connect to other networks



IPsec VPN creates dynamic connections



IPsec VPN protocol



Creates a VPN “tunnel” connection



Uses Cloud Router to make the connection dynamic

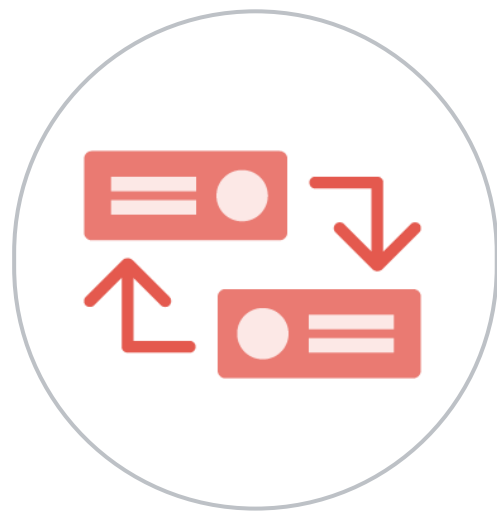


Lets other networks and Google VPC exchange route information over the VPN using the Border Gateway Protocol



Not always the best option because of security concerns or bandwidth reliability

Direct Peering routes traffic through a Google PoP



Direct Peering



Puts a router in the same public datacenter as a Google point of presence (PoP)



Uses a router to exchange traffic between networks



Connects to more than 100 Google points of presence around the world

Carrier Peering allows access to Google Workspace



Carrier Peering



Puts a router in the same public datacenter as a Google point of presence (PoP)



Uses a router to exchange traffic between networks



Connects to more than 100 Google points of presence around the world

Dedicated Interconnect is a direct connection to Google



Dedicated Interconnect



Good solution for getting the highest uptimes for interconnection



Allows for one or more direct, private connections to Google



Can be covered by up to a 99.99% SLA if connections have topologies that meet Google's specifications



Connections can be backed up by a VPN for even greater reliability

Partner Interconnect links your on-prem with your VPC



Partner
Interconnect



Provides connectivity between an on-premises network and a VPC network through a supported service provider



Can be configured to support mission-critical services or applications that can tolerate some downtime



Can be covered by up to a 99.99% SLA if connections have topologies that meet Google's specifications

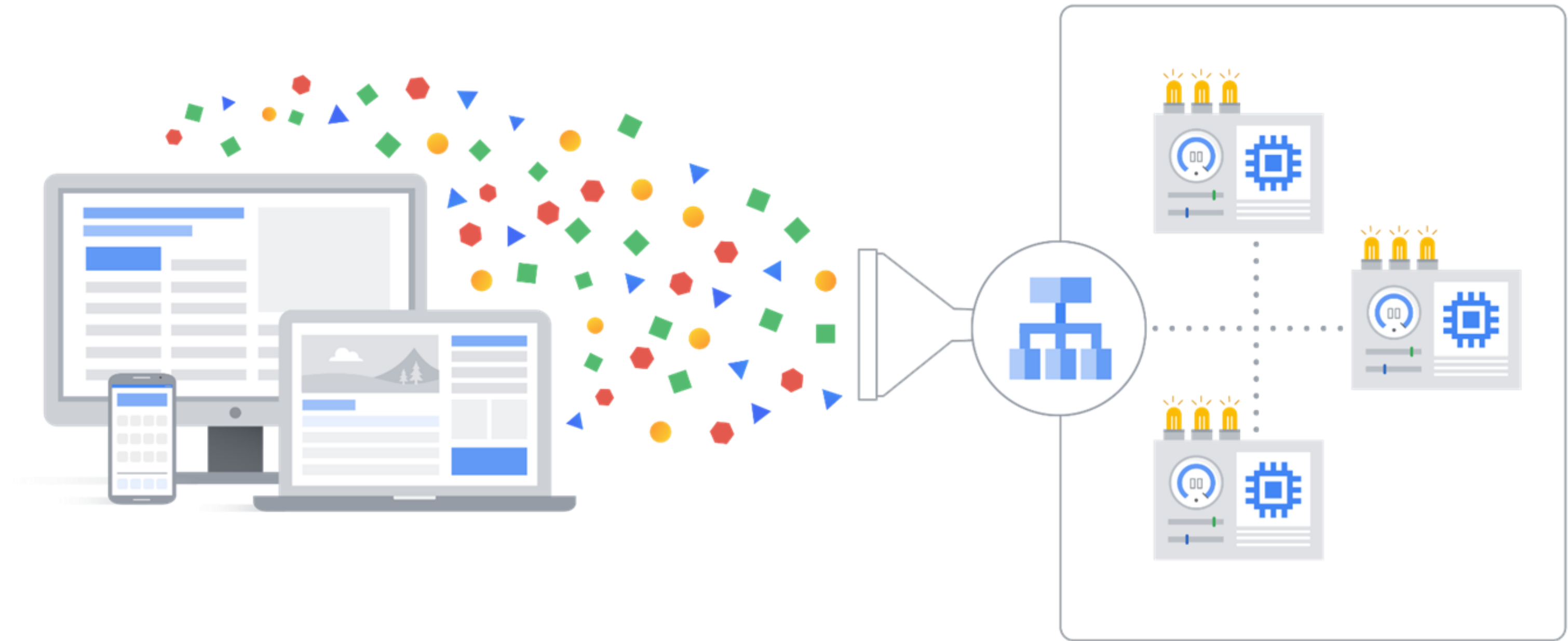


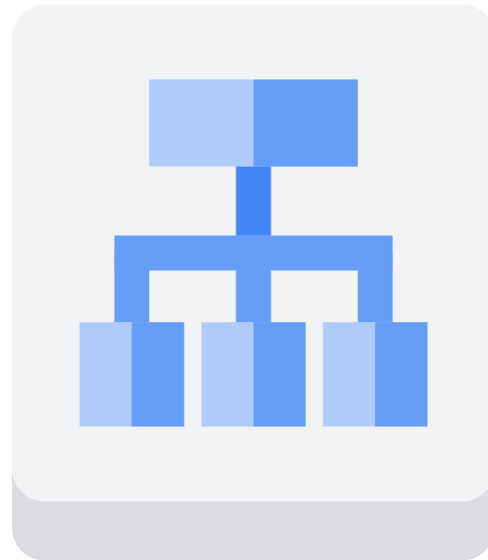
Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options**
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13 Summary

Load balancing distributes traffic across instances





Cloud Load Balancing



Fully distributed, software-defined, managed service



You can put Cloud Load Balancing in front of all of your traffic:



HTTP(S)



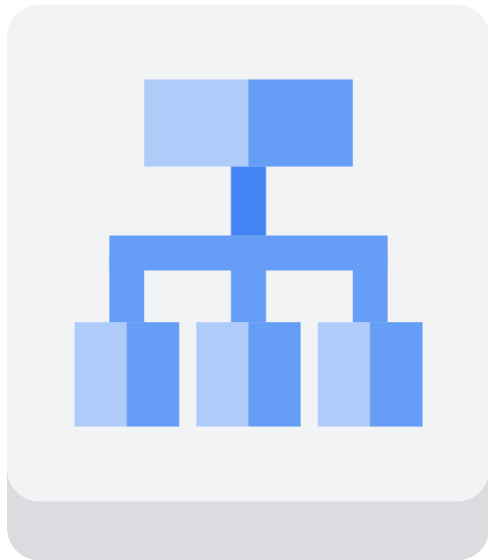
TCP traffic



SSL traffic



UDP traffic



Cloud Load Balancing

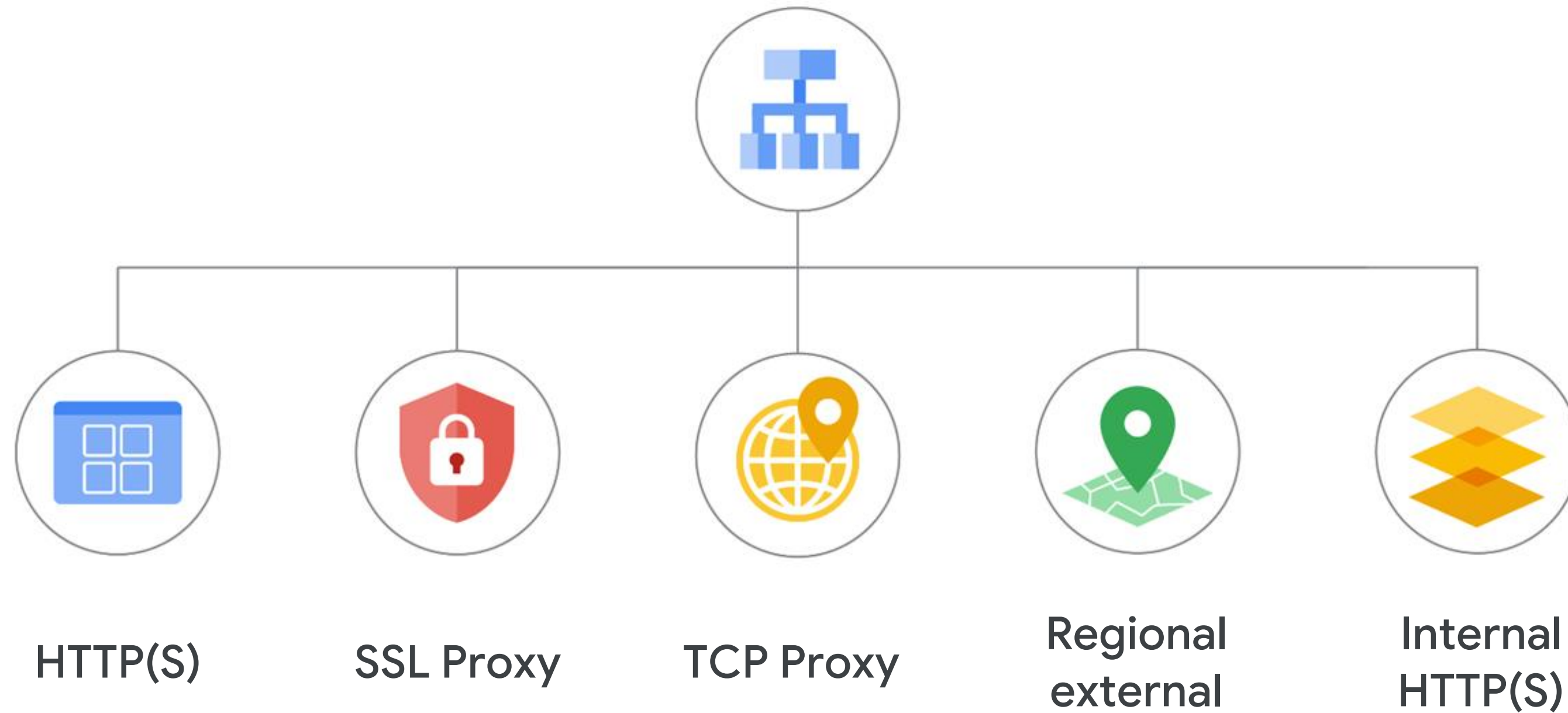


Provides single as well as cross-region load balancing, including automatic multi-region failover



No “pre-warming” is required for anticipated spikes in traffic

Load balancing options





Agenda



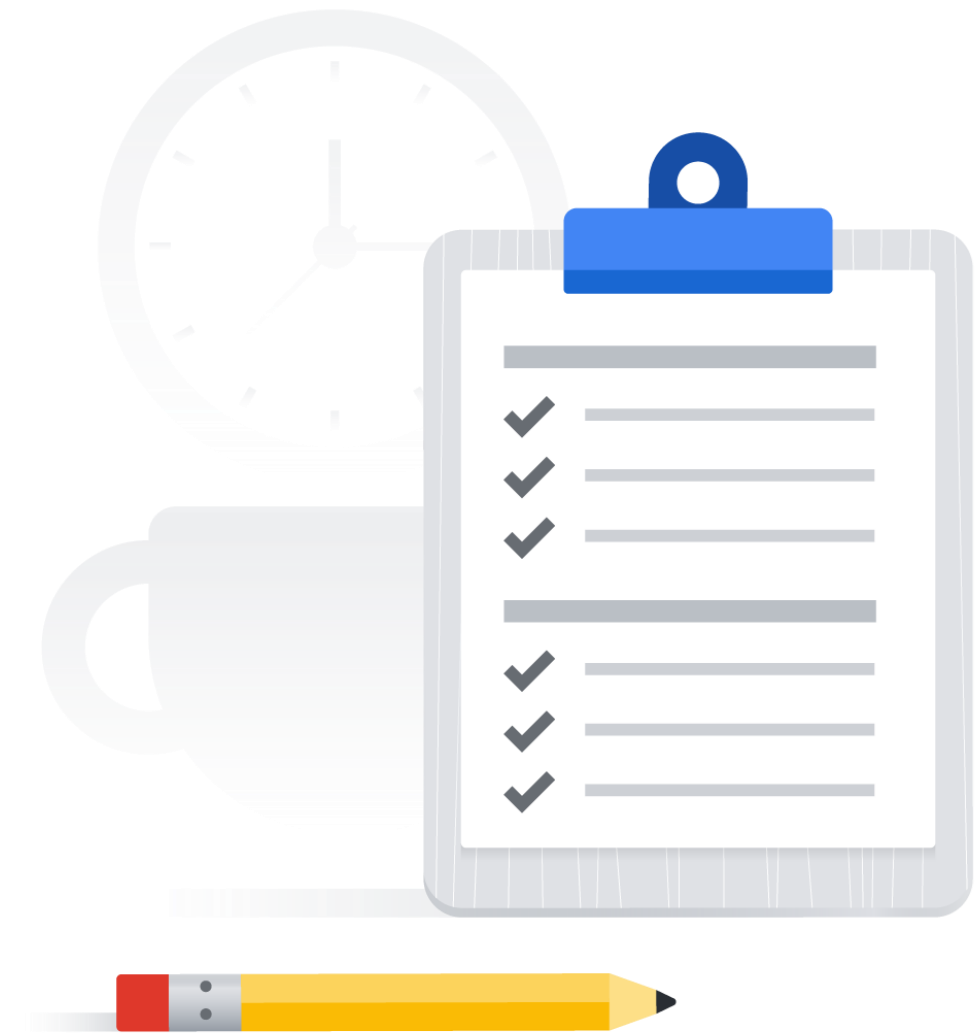
- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor**
- 12 Quiz
- 13 Summary

Lab: HTTP Load Balancer with Google Cloud Armor

 Hands-on lab

During this lab, you'll:

1. Create HTTP and health check firewall rules.
2. Configure two instance templates.
3. Create two managed instance groups.
4. Configure an HTTP load balancer with IPv4 and IPv6.
5. Stress test an HTTP load balancer.
6. Blocklist an IP address to restrict access to an HTTP load balancer.





Agenda



- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz**
- 13 Summary

Knowledge check

Select the true statement.

- A. VPCs are global and subnets are regional.
- B. VPCs are regional and subnets are zonal.
- C. VPCs are regional. Subnets are not used in Google Cloud.
- D. Both VPCs and subnets are global.



Knowledge check

Select the true statement.

- A. VPCs are global and subnets are regional.
- B. VPCs are regional and subnets are zonal.
- C. VPCs are regional. Subnets are not used in Google Cloud.
- D. Both VPCs and subnets are global.



Knowledge check

What term describes where the Google network is connected to the rest of the internet?

- A. Zone
- B. Point of presence (PoP)
- C. IP address
- D. Region



Knowledge check

What term describes where the Google network is connected to the rest of the internet?

A. Zone

B. Point of presence (PoP)

C. IP address

D. Region



Knowledge check

What is used to forward traffic from one instance to another within the same network, across subnetworks, or even between Google Cloud zones, and doesn't require an external IP address?

- A. Load balancer
- B. Subnet
- C. Firewall
- D. Routing table



Knowledge check

What is used to forward traffic from one instance to another within the same network, across subnetworks, or even between Google Cloud zones, and doesn't require an external IP address?

- A. Load balancer
- B. Subnet
- C. Firewall
- D. Routing table



Knowledge check

Which connectivity solution provides a direct connection to Google?

- A. Partner Interconnect
- B. Dedicated Interconnect
- C. Direct Peering
- D. Carrier Peering



Knowledge check

Which connectivity solution provides a direct connection to Google?

A. Partner Interconnect

B. Dedicated Interconnect

C. Direct Peering

D. Carrier Peering



Knowledge check

Which option would you choose if you required cross-regional load balancing for a web application?

- A. TCP Proxy Load Balancing
- B. SSL Proxy Load Balancing
- C. HTTP(S) Load Balancing
- D. Internal HTTP(S) Load Balancing



Knowledge check

Which option would you choose if you required cross-regional load balancing for a web application?

- A. TCP Proxy Load Balancing
- B. SSL Proxy Load Balancing
- C. HTTP(S) Load Balancing
- D. Internal HTTP(S) Load Balancing



Agenda

- 01 Networking in the cloud
- 02 Virtual Private Clouds (VPCs)
- 03 The basics of public and internal IP addresses
- 04 The Google Cloud network
- 05 Routes and firewall rules in the cloud
- 06 Multiple VPC networks
- 07 Lab: Multiple VPC Networks
- 08 Lab: VPC Networks - Controlling Access
- 09 Building hybrid clouds
- 10 Load balancing options
- 11 Lab: HTTP Load Balancer with Google Cloud Armor
- 12 Quiz
- 13** Summary

Summary

- ✓ Explored Virtual Private Clouds (VPCs).
- ✓ Examined Google's network architecture.
- ✓ Learned how to use multiple VPC networks.
- ✓ Explored options to build hybrid clouds.
- ✓ Examined load balancing options.



